

Section : Common

Q.1 The solution of the differential equation $xdy - ydx = 0$ represents

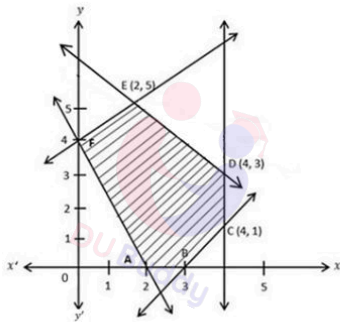
1. straight line passing through origin
2. a circle whose centre is at origin
3. a rectangular hyperbola
4. parabola whose vertex is at origin

Options 1. 1

2. 2
3. 3
4. 4

Q.2

With respect to the following shaded feasible region (ABCDEFA), the maximum value of the objective function $z = 3x + 4y - 2$ is at point(s):



1. Point B and D
2. Point F only
3. Point A and C
4. Point E only

Options 1. 1

2. 2
3. 3
4. 4

Q.3 $\int_1^2 \frac{1}{x(x+1)} dx, x > 0$ equals

1. $\log\left(\frac{2}{3}\right)$
2. $\log\left(\frac{4}{3}\right)$
3. 0
4. $\log\left(\frac{3}{4}\right)$

Options 1. 1

2. 2
3. 3
4. 4

Q.4

In the following differential equation $\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 = 2x^2 \log\left(\frac{d^2y}{dx^2}\right)$ order and degree is:

1. order 2, degree 2
2. order 2, degree not defined
3. order not defined, degree 2
4. order 1, degree 2

Options 1. 1

2. 2
3. 3

Q.5 $\int \frac{(x-1)e^x}{x^2} dx, x > 0$ equals (where C is an arbitrary constant)

1. $xe^x + C$
2. $\frac{e^x}{x} + C$
3. $\frac{e^{-x}}{x^2} + C$
4. $e^x + \frac{1}{x} + C$

Options 1. 1

2. 2
3. 3
4. 4

Q.6 If P and Q are non-singular square matrices of the same order, then $(PQ^{-1})^{-1}$ equals

1. PQ
2. $P^{-1}Q$
3. $P^{-1}Q^{-1}$
4. QP^{-1}

Options 1. 1

2. 2
3. 3
4. 4

Q.7 If the objective function $z = 4x + 3y$ has maximum value on a line joining points $(3, a)$ and $(b, 2)$ where $a > 0, b > 0$ such that $a - b = 2$, then the maximum value of z is:

1. 46
2. 54
3. 56
4. 92

Options 1. 1

2. 2
3. 3
4. 4

Q.8 If the random variable X has the following probability distribution:

X 0 1 2 otherwise

P(X) k 3k 5k 0

$X=0, P(X)=k; X=1, P(X)=3k; X=2, P(X)=5k; \text{otherwise}, P(X)=0$

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \textbf{List-II}

(A) k (I) $\frac{13}{9}$

(B) $E(X)$ (II) $\frac{4}{9}$

(C) $P(X \leq 1)$ (III) $\frac{8}{9}$

(D) $P(1 \leq X \leq 2)$ (IV) $\frac{1}{9}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
2. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
3. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.9 $\frac{d}{dx} (e^{2\log_e x^3})$ equals

1. $30x^4$
2. $6x^5$
3. $5x^6$
4. $3x^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.10 The area of the region bounded by the curve $y = x + 1$, x -axis and the lines $x = 2$ and $x = 3$ is

1. $\frac{9}{2}$ sq. units
2. $\frac{7}{2}$ sq. units
3. $\frac{13}{2}$ sq. units
4. $\frac{5}{2}$ sq. units

Options 1. 1

2. 2
3. 3
4. 4

Q.11 If $A = \begin{bmatrix} x+z & 2 & -3 \\ x & 0 & 4 \\ 3 & x-y & 0 \end{bmatrix}$ is a skew-symmetric matrix, then which of the following are true?

- (A) $y > z > x$
- (B) $x > y$
- (C) $x + y + z > 0$
- (D) $z > x$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (D) only
2. (B), (C) and (D) only
3. (B) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.12 The function $f(x) = x^3 + 3x^2 + 4x + 4$, $x \in \mathbb{R}$ (set of real numbers) :

1. is increasing on $\mathbb{R}$
2. is decreasing on $\mathbb{R}$
3. is decreasing on $(-\infty, 0)$
4. is neither increasing nor decreasing on $(0, \infty)$

Options 1. 1

2. 2

3. 3
4. 4

Q.13

If $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix}$ then

1. $AB = 0, BA \neq 0$
2. $AB = 0, BA = 0$
3. $A^2 \neq A$
4. $B^2 \neq 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.14

If $\begin{bmatrix} 1 & 0 & 0 \\ 0 & y+1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2x \\ -2 \\ z-3 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix}$ then $x + y + z$ is

1. 10
2. 4
3. 5
4. 7

Options 1. 1

2. 2
3. 3
4. 4

Q.15

The maximum value of the function $f(x) = x^2(60 - x)$ in $[20, 80]$ is:

1. 16000
2. 32000
3. 64000
4. 128000

Options 1. 1

2. 2
3. 3
4. 4

Section : Applied Mathematics

Q.16

Inlet pipe A can fill a tank in 30 minutes and outlet pipes B and C can empty the tank in 2 hours each.

If all three pipes operate together, the tank will be filled in:

1. 1 hour
2. 2 hours
3. 3 hours
4. The tank will not be filled

Options 1. 1

2. 2
3. 3
4. 4

Q.17

A boat covers a distance of 24 km upstream and returns to the same point in a total time of 4 hours.

If the speed of the boat in downstream is twice its speed in upstream, then the speed of boat in upstream is :

1. 3 km/h
2. 6 km/h
3. 9 km/h
4. 18 km/h

Options 1. 1

2. 2

3. 3
4. 4

Q.18 Vatsala buys a car for Rs.7,00,000 and pays upfront Rs.2,50,000 through her credit card.
The balance is to be paid in 5 years by equal monthly installments at an interest of 7\% per annum as reducing balance.
The EMI to be paid by Vatsala will be :- given $(1.0058)^{-60} = 0.7068$

1. Rs. 3692.7
2. Rs. 13,847.2
3. Rs. 5,744.2
4. Rs. 8901.7

- Options 1. 1
2. 2
3. 3
4. 4

Q.19 The value of $\left| \frac{\log_5 10}{2} - \frac{2}{\log_{10} 5} \right|$ is

1. -3
2. -7/2
3. 1
4. 0

- Options 1. 1
2. 2
3. 3
4. 4

Q.20 Consider the following test:

$$H_0 : \mu \leq 12$$

$$H_a : \mu > 12$$

A sample of 36 provided a sample mean $\bar{x} = 16$ and a sample standard deviation $S = 4.2$

Then the value of the t-test statistic is:

1. 5.32
2. 2.67
3. 34.26
4. 5.71

- Options 1. 1
2. 2
3. 3
4. 4

Q.21 Consider the following data

Year(X)	2010	2011	2012	2013	2014
Profit (Rs. in thousands)(y)	10	12	14	16	13

The equation of straight line trend by method of least square for the above data is given by

1. $y = 2x - 26$
2. $y = 2x + 26$
3. $y = x - 1999$
4. $y = x - 13$

- Options 1. 1
2. 2
3. 3
4. 4

Q.22 The solution of $\frac{7x+12}{x-9} < 4, x \neq 9$ is:

- (1) $\{x : 9 < x < 16, x \in \mathbb{R}\}$
- (2) $\{x : -16 < x < -9, x \in \mathbb{R}\}$

- (3) $\{x : -16 < x < 9, x \in \mathbb{R}\}$
 (4) $\{x : -9 < x < -16, x \in \mathbb{R}\}$

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.23

If $\begin{bmatrix} a-b & 0 & 0 \\ 0 & b-c & 0 \\ 0 & 0 & c-2 \end{bmatrix}$ is a scalar matrix such that $a + b + c = 0$, then, which of the following are TRUE?

- (A) $a = 0$
 (B) $b = 0$
 (C) $a = 1$
 (D) $c = 1$

Choose the correct answer from the options given below:

1. (A) and (C) only
 2. (B) and (D) only
 3. (C) and (D) only
 4. (A), (C) and (D) only

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.24

The least non-negative remainder when 2^{75} is divided by 5 will be:-

1. 1
 2. 2
 3. 3
 4. 4

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.25

The probability that in a year of the 22nd century chosen at random, there will be 53 Sundays is:

1. $\frac{3}{28}$
 2. $\frac{2}{7}$
 3. $\frac{5}{28}$
 4. $\frac{1}{7}$

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.26

A square board of side 36cm is made into a box without top by cutting a square from each corner and folding up the flaps to form a box then maximum volume of the box is

1. 2856 cm^3
 2. 3256 cm^3
 3. 3656 cm^3
 4. 3456 cm^3

Options 1. 1
 2. 2
 3. 3

Q.27 The volume of spherical balloon is increasing at the rate of $4 \text{ cm}^3/\text{sec}$.
The rate of increase of its surface area, when the radius is 3 cm will be :-

1. $2.5 \text{ cm}^2/\text{sec}$
2. $2.66 \text{ cm}^2/\text{sec}$
3. $2.61 \text{ cm}^2/\text{sec}$
4. $2.59 \text{ cm}^2/\text{sec}$

Options 1. 1

2. 2
3. 3
4. 4

Q.28 With reference to sampling, which of the following are correct?

- (A) Simple random sampling is probability sampling
- (B) Snow-ball sampling is non-probability sampling
- (C) Stratified sampling is probability sampling
- (D) Cluster sampling is non-probability sampling

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (D) and (C) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.29 The amount to be deposited at the end of every 6 months to accumulate Rs. 50,000 in 8 years, if money is worth 6% p.a. compounded semiannually, is:

(Given $(1.03)^{16} = 1.6047$)

- (1) Rs. 3432.53
- (2) Rs. 2783.08
- (3) Rs. 2480.57
- (4) Rs. 2149.93

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Curd is at 80°F , five minutes later it came down at 60°F .

After another 5 minutes, its temperature became 50°F .

Given that the rate of change of temperature is proportional to $(T - S)$, where S is temperature of surroundings and T is temperature of the curd at any time t.

Then the temperature of the surroundings is:

1. 20°F
2. 40°F
3. 30°F
4. 50°F

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The interval(s), where the function $f(x) = \begin{cases} \frac{1-e^x}{e^{2x}-1} : x \neq 0 \\ -\frac{1}{2} : x = 0 \end{cases}$ is increasing, is/are:

1. $(1, \infty)$
2. $(-\infty, 1)$
3. $(-\infty, \infty)$

4. $(-\infty, \infty) - \{0\}$

Options 1. 1

2. 2

3. 3

4. 4

Q.32 Anisha invested Rs.20000 in a mutual fund in the year 2016, which increased to Rs.36000 in the year 2024. The percentage compounded annual growth rate (CAGR) of her investment is:

(Given: $(1.8)^{1/8} = 1.076$)

1. 13.8% p.a

2. 7.6% p.a

3. 8.6% p.a

4. 10.76% p.a

Options 1. 1

2. 2

3. 3

4. 4

Q.33 Match List-I with List-II

List I

List II

Time Series Component Example

(A) Secular Variation (I) Pandemic

(B) Seasonal Variation (II) Recession in business

(C) Cyclic Variation (III) Monthly sale of woolen clothes

(D) Irregular Variation (IV) Data regarding National income

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

4. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.34 The demand function P for maximising a profit monopolist is given by $P = 274 - x^2$, while the marginal cost is $4 + 3x$ for x units of commodity.

The consumer surplus is

1. 274

2. 1737

3. 193

4. 486

Options 1. 1

2. 2

3. 3

4. 4

Q.35 In a Binomial distribution, the probability of getting a success is $\frac{3}{4}$

and the variance is $\frac{3}{8}$ then the probability of no success is:

1. $\frac{1}{4}$

2. $\frac{3}{4}$

3. $\frac{1}{16}$

4. $\frac{3}{16}$

- Options 1. 1
2. 2
3. 3
4. 4

Q.36

The integral $\int \frac{2 dx}{e^{2x} - 1}$ is equal to:

1. $\log |e^{2x} - 1| + C$: C is an arbitrary constant
2. $\log \left| \frac{e^{2x} - 1}{e^{2x} + 1} \right| + C$: C is an arbitrary constant
3. $\log \left| \frac{e^{2x}}{e^{2x} - 1} \right| + C$: C is an arbitrary constant
4. $\log \left| \frac{e^{2x} - 1}{e^{2x}} \right| + C$: C is an arbitrary constant

- Options 1. 1
2. 2
3. 3
4. 4

Q.37 Match List-I with List-II

List I

Matrix/Equations

List II

Values

(A) $\begin{bmatrix} 2x+1 & 3y \\ 0 & y^2-5y \end{bmatrix} = \begin{bmatrix} x+3 & y^2+2 \\ 0 & -6 \end{bmatrix}$ (I) $x = 2, y = -1$

(B) $\begin{bmatrix} 1 & 2 & -1 \\ x & 0 & 3 \\ y & 3 & 4 \end{bmatrix}$ is symmetric (II) $x = 2, y = 2$

(C) $[x \ 1] \begin{bmatrix} 1 & 0 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 0 & y \end{bmatrix} = 0$ (III) $x = -2, y = 2$

(D) $\begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -1 & y/2 \end{bmatrix}$ (IV) $x = 2, y = 0$

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

- Options 1. 1
2. 2
3. 3
4. 4

Q.38 Match List-I with List-II

List I

List II

Terms

Definitions

(A) POPULATION (I) Measurable characteristics of the population such as mean, variance, standard deviation etc. of population

(B) SAMPLE (II) Measurable characteristics of the sample such as mean, variance, standard deviation etc. of a sample

(C) PARAMETER (III) Finite set of statistical individuals drawn from a population for investigation

(D) STATISTIC (IV) Collection of objects having the same characteristics

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

3. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.39

Which of the following are the properties of Normal Distribution function $f(x)$ and Normal probability curve:

(A) The probability of success remains the same in each trial and the number of trials is small in number.

(B) The curve is bell shaped and is symmetrical about the mean.

(C) If set of n trials are repeated N times, then frequency $f(r)$ of r successes is given by

$$f(r) = N.P(r) = Ne^{-m} \frac{m^r}{r!}, r = 0, 1, 2 \dots$$

(D) As x increases numerically, $f(x)$ decreases rapidly and the maximum value of $f(x)$ occurs at $x = \mu$ (mean).

Choose the correct answer from the options given below:

1. (A), (B) and (C) only

2. (B), (C) and (D) only

3. (B) and (D) only

4. (A) only

Options 1. 1

2. 2

3. 3

4. 4

Q.40

The point estimate of the population standard deviation as per the below mentioned data from a simple random sample

6, 10, 15, 12, 9, 8 will be :-

1. 3.612

2. 3.126

3. 3.162

4. 3.216

Options 1. 1

2. 2

3. 3

4. 4

Q.41

If a matrix $A = \begin{bmatrix} 5 & -8 \\ -3 & 5 \end{bmatrix}$ then which of the following is / are TRUE?

(A) $|A| = 1$

(B) A is a singular matrix.

(C) $-2A = \begin{bmatrix} 10 & -16 \\ -6 & 10 \end{bmatrix}$

(D) $AI = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$ I is an identity matrix of order 2.

Choose the correct answer from the options given below:

1. (A) only

2. (A) and (C) only

3. (A), (B) and (C) only

4. (A), (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.42

The random variable X has the following probability distribution

X 0 1 2 3

P(X) a a b b

such that $E(x^2) = 2E(x)$, then the value of b is:

1. $\frac{1}{8}$
2. $\frac{1}{4}$
3. $\frac{1}{2}$
4. $\frac{3}{8}$

Options 1. 1
2. 2
3. 3
4. 4

Q.43 At what rate will the present value of a perpetuity of Rs.1000 payable at the end of each quarter be Rs.50000?

1. 5\% p.a
2. 8\% p.a
3. 2\% p.a
4. 4\% p.a

Options 1. 1
2. 2
3. 3
4. 4

Q.44

The annual depreciation of an asset is independent of:-

1. Useful life of the asset.
2. Original value / cost of the asset.
3. Profit gained from the asset.
4. Scrap value of the asset.

Options 1. 1
2. 2
3. 3
4. 4

Q.45 If A and B are two matrices of order 2×2 such that A is a symmetric matrix and B is a skew-symmetric matrix, then:

1. A^2 is also a symmetric matrix
2. B^2 is also a skew-symmetric matrix
3. $A + B$ is an identity matrix
4. $A - B$ is a null matrix

Options 1. 1
2. 2
3. 3
4. 4

Q.46 A man wishes to ensure that he gets Rs. 75,000/- at the end of each year indefinitely.

The amount that he invest now to produce the desired cash flow, if money is worth 2.5\% compounded annually is:

1. Rs. 30,66,337
2. Rs. 30,00,000
3. Rs. 30,45,339
4. Rs. 30,66,373

Options 1. 1
2. 2
3. 3
4. 4

Q.47 From a container full of orange juice, 7.5 liters was drawn out and replaced by soda water.

This process is repeated 5 more time.

The ratio of quantity of orange juice and soda water left in the container is 4:5.

How much liter of orange juice did the container originally had?

(Use $(0.44)^{1/6} = 0.802$)

1. 20.44 litres
2. 30.802 litres
3. 37.9 litres
4. 17.5 litres

Options 1. 1

2. 2
3. 3
4. 4

Q.48 For the objective function $Z = 4x + 6y$ subject to the constraints

$3x + 2y \geq 5$, $7x + 2y \leq 9$, $x \geq 0$, $y \geq 0$,

the maximum value of Z occurs at (a, b)

and the minimum value of Z occurs at (p, q)

then the value of $\frac{a}{p} + \frac{b}{q}$ is:

1. 2.5
2. 4.5
3. 5.5
4. 3.5

Options 1. 1

2. 2
3. 3
4. 4

Q.49 If the corner points of bounded feasible region for an LPP are $(0, 2)$, $(3, 0)$, $(6, 0)$, $(6, 8)$ and $(0, 5)$

then the minimum value of the objective function $f = 4x + 6y$ occur at

1. $(3, 0)$ only
2. $(0, 2)$ only
3. Every point on the line segment joining the points $(0, 2)$ and $(3, 0)$
4. Only the mid point of the line segment joining the points $(0, 2)$ and $(3, 0)$

Options 1. 1

2. 2
3. 3
4. 4

Q.50 In a game, A can give 36 points to B, A can give 42 point to C, B can give 10 points to C.

How many points make the game ?

1. 360 points
2. 88 points
3. 90 points
4. 350 points

Options 1. 1

2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51 If $\vec{a}$, $\vec{b}$ and $\sqrt{3}\vec{a} + \vec{b}$ are unit vectors, then the angle between $\vec{a}$ and $\vec{b}$ is:

1. $\frac{\pi}{2}$
2. $\frac{\pi}{3}$
3. $\frac{\pi}{6}$

4. $\frac{5\pi}{6}$

- Options 1. 1
2. 2
3. 3
4. 4

Q.52 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = [x]$, where $[x]$ denotes the greatest integer less than or equal to x . Then which of the following statements are **\textbf{correct}**?

- (A) f is one-one but not onto
(B) f is not onto
(C) f is not one-one
(D) f is one-one and onto

Choose the **\textbf{correct}** answer from the options given below:

1. (B) and (C) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (C) and (D) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.53 The value of $\int_0^{\pi/2} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx$ is

1. $\pi/2$
2. $\pi/3$
3. $\pi/4$
4. $\pi/6$

- Options 1. 1
2. 2
3. 3
4. 4

Q.54 A and B throw a die alternatively till one of them gets 3 or 6 and wins the game. If B starts the game, then the probability of winning the game by A is

1. $2/5$
2. $3/5$
3. $6/11$
4. $5/11$

- Options 1. 1
2. 2
3. 3
4. 4

Q.55 The area (in sq. units) of the region bounded by the curve $x^2 = 250y$, $y = 0$ and $x = 50$ is

1. $\frac{1000\sqrt{2}}{3}$
2. 1000
3. $\frac{4}{3}$
4. $\frac{500}{3}$

- Options 1. 1
2. 2
3. 3
4. 4

Q.56 The edge of a cube is increasing at a rate of 7 cm/s. The rate of change of area of the cube when its side is 3 cm is:

1. $525 \text{ cm}^2/\text{s}$
2. $225 \text{ cm}^2/\text{s}$
3. $255 \text{ cm}^2/\text{s}$
4. $252 \text{ cm}^2/\text{s}$

Options

1. 1
2. 2
3. 3
4. 4

Q.57 The corner points of the bounded feasible region of the LPP: Maximize $Z = x + y$ subject to constraints $2x + 5y \leq 100$, $8x + 5y \leq 200$, $x \geq 0$, $y \geq 0$ are

1. $(0, 0)$, $(25, 0)$, $(0, 20)$, $\left(\frac{50}{3}, \frac{40}{3}\right)$
2. $(25, 0)$, $(20, 0)$, $\left(\frac{50}{3}, \frac{40}{3}\right)$, $(50, 0)$
3. $(0, 20)$, $(0, 40)$, $(50, 0)$, $(25, 0)$
4. $(0, 0)$, $(50, 0)$, $(0, 20)$, $\left(\frac{50}{3}, \frac{40}{3}\right)$

Options

1. 1
2. 2
3. 3
4. 4

Q.58 The function $f(x) = \begin{cases} \frac{\sin 2x}{x} + \cos x & , \text{ if } x \neq 0 \\ K & , \text{ if } x = 0 \end{cases}$ is continuous at $x = 0$, then the value of K is:

1. 1
2. 2
3. 0
4. 3

Options

1. 1
2. 2
3. 3
4. 4

Q.59 If the points $(-1, -1, 2)$, $(2, m, 5)$ and $(3, 11, 6)$ are collinear, then m equals

1. 4
2. 8
3. 0
4. -2

Options

1. 1
2. 2
3. 3
4. 4

Q.60 The corner points of a bounded feasible region determined by the following system of linear inequalities $x + 3y \leq 60, x + y \geq 10, x \leq y, x \geq 0, y \geq 0$ are $(0, 10), (5, 5), (15, 15)$ and $(0, 20)$. Let $z = 2px + qy, p, q > 0$. If maximum of z occurs at both $(15, 15)$ and $(0, 20)$, then the relation between p and q is

1. $3p = 2q + 1$
2. $2p = 4q$
3. $6p = q$
4. $2p = 3q$

Options 1. 1

2. 2
3. 3
4. 4

Q.61 If $e^x + e^y = e^{x+y}$, then $\frac{dy}{dx}$ equals

1. $-e^{y-x}$
2. e^{y-x}
3. e^{x-y}
4. $-e^{x-y}$

Options 1. 1

2. 2
3. 3
4. 4

Q.62 If A and B are two distinct events such that $P(A|B) = P(B|A)$, then which of the following is /are possible?

- (A) $A = B$
- (B) $P(A) = P(B)$
- (C) $A \subset B$ but $A \neq B$
- (D) $A \cap B = \phi$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (B) only
2. (B) only
3. (B) and (D) only
4. (A), (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.63 Which of the following statements are \textbf{true} ?

- (A) The vector equation of the line through the point $(5, 2, -4)$ and parallel to the vector $3\hat{i} + 2\hat{j} - 8\hat{k}$ is $\vec{r} = (5\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(3\hat{i} + 2\hat{j} - 8\hat{k})$
- (B) Vector form of the equation of line $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$ is $\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} + 2\hat{k})$
- (C) The direction cosines of z-axis are $(1, 1, 0)$.
- (D) If a line has direction ratios $2, -1, -2$, then its direction cosines are $-2/3, -1/3, -2/3$.

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A) and (B) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.64

If $y = -4$ is a root of $\begin{vmatrix} y & 2 & 3 \\ 1 & y & 1 \\ 3 & 2 & y \end{vmatrix} = 0$, then the product of the other two roots is

1. -5
2. 4
3. 0
4. 3

Options 1. 1

2. 2
3. 3
4. 4

Q.65

The value of $\begin{vmatrix} 2^x & 1 & 6^x \\ 4^x & 1 & 3^x \\ 2^x & 1 & 6^x \end{vmatrix}$, where $x \neq 0$; is:

1. 0
2. $2^x + 3^x$
3. $3^x + 4^x$
4. $4^x + 6^x$

Options 1. 1

2. 2
3. 3
4. 4

Q.66

Arrange the principal values of the following functions in ascending order

- (A) $\operatorname{cosec}^{-1}(2)$
- (B) $\tan^{-1}(-\sqrt{3})$
- (C) $\tan^{-1}(1)$
- (D) $\tan^{-1}\left(\cos \frac{3\pi}{2}\right)$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (D), (B), (C)
3. (B), (D), (A), (C)
4. (D), (A), (B), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.67

Let $A = \{1, 2, 3\}$. The number of equivalence relations containing $(1, 3)$ is

1. 1
2. 2
3. 3
4. 5

Options 1. 1

2. 2
3. 3
4. 4

Q.68

Derivative of x^x with respect to $x \log x$ is

1. x
2. $x \log x$
3. x^x
4. $\log x$

Options 1. 1

2. 2
3. 3
4. 4

Q.69

The value of p so that the lines $\frac{x-1}{-3} = \frac{2y-2}{2p} = \frac{z-3}{2}$ and $\frac{x-1}{-3p} = \frac{y-1}{4} = \frac{6-z}{5}$ are at right angles is

1. $-\frac{10}{17}$
2. $-\frac{10}{13}$
3. $\frac{10}{13}$
4. $\frac{10}{17}$

Options 1. 1

2. 2
3. 3
4. 4

Q.70

If $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$, then which of the following statement(s) is/are \textbf{correct}?

- (A) A is singular matrix
- (B) $|3A| = 135$
- (C) $|\text{adj } A| = 125$
- (D) $|A^{-1}| = \frac{1}{5}$

Choose the \textbf{correct} answer from the options given below:

1. (B) and (D) only
2. (A) and (C) only
3. (B) and (C) only
4. (A) only

Options 1. 1

2. 2
3. 3
4. 4

Q.71

Area (in sq. units) of the region bounded by the curves $y = -1, y = 2, x = y^3$ and $x = 0$ is

1. 4
2. $\frac{17}{4}$
3. 8
4. $\frac{19}{4}$

Options 1. 1

2. 2
3. 3
4. 4

Q.72

$\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$ is equal to

1. $\frac{e^x}{1+x^2} + C$: C is an arbitrary constant
2. $\frac{e^x}{(1+x^2)^2} + C$: C is an arbitrary constant
3. $\frac{-e^x}{(1+x^2)^2} + C$: C is an arbitrary constant
4. $\frac{-e^x}{1+x^2} + C$: C is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.73 If it is given that at $x = 1$, the function $f(x) = x^4 - 62x^2 + 2ax + 8$ attains its maximum value on the interval $[0, 2]$, then the value of a is:

1. 120

2. 60

3. 0

4. -60

Options 1. 1

2. 2

3. 3

4. 4

Q.74 If $x = a \sec^3 \theta$, $y = a \tan^3 \theta$, then $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$ is

1. $\frac{\sqrt{3}}{2}$

2. $\frac{1}{2}$

3. 0

4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.75 Let $A = [a_{ij}]_{3 \times 3}$ be a matrix, defined by $a_{ij} = \begin{cases} 2i + 3j & , i < j \\ 6 & , i = j \\ 3i - 2j & , i > j \end{cases}$. The number of elements in A which are greater than 6, is

1. 6

2. 5

3. 4

4. 3

Options 1. 1

2. 2

3. 3

4. 4

Q.76 For two matrices $A = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B^T = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, $A - B$ equals

1. $\begin{bmatrix} 3 & 3 \\ -2 & 0 \\ 0 & -3 \end{bmatrix}$

2. $\begin{bmatrix} 4 & 3 \\ 0 & -3 \\ -1 & -2 \end{bmatrix}$

3. $\begin{bmatrix} 4 & 3 \\ 2 & -3 \\ 0 & -4 \end{bmatrix}$

4. $\begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.77 Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}

\textbf{List-II}

\textbf{Differential Equations}

\textbf{Order and degree}

(A) $ydx + x \log(y/x)dy - 2xdy = 0$

(I) Order : 2, degree:1

(B) $\left(\frac{d^3y}{dx^3}\right)^2 + 3\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^4 = y^2$

(II) Order :1, degree:1

(C) $\frac{dy}{dx} + \log\left(\frac{dy}{dx}\right) + x = y$

(III) Order : 3, degree:2

(D) $\left(\frac{ds}{dt}\right)^4 + 2s\frac{d^2s}{dt^2} = 0$

(IV) Order : 1, degree: Not defined

Choose the \textbf{correct} answer from the options given below:

- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
- (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
- (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.78 If $f(x)$ and $g(x)$ are continuous functions in $[0, a]$ such that $f(x) = f(a - x)$ and $g(x) + g(a - x) = a$ then

$\int_0^a f(x)g(x)dx =$

1. $a \int_0^a f(x)dx$

2. $a \int_0^a g(x)dx$

3. $\frac{a}{2} \int_0^a f(x)dx$

4. $\frac{a}{2} \int_0^a g(x)dx$

Options 1. 1

2. 2

3. 3

4. 4

Q.79 If the events A and B are independent, then which of the following statements are \textbf{true}?

(A) $P(A' \cap B) = [1 - P(A)]P(B)$

(B) A and B are mutually exclusive

(C) $P(A) = P(B)$

(D) $P(A' \cap B') = [1 - P(A)][1 - P(B)]$

Choose the \textbf{correct} answer from the options given below:

- (A), (B) and (C) only
- (B), (C) and (D) only
- (C) and (D) only
- (A) and (D) only

Options 1. 1

2. 2

3. 3

Q.80

A line passes through the point with position vector $2\hat{i} - \hat{j} + 4\hat{k}$ and is in the direction of the vector $\hat{i} + \hat{j} - 2\hat{k}$. The equation of the line in Cartesian form is:

1. $\frac{x-2}{1} = \frac{y+1}{1} = \frac{4-z}{2}$
2. $\frac{x+2}{1} = \frac{y-1}{1} = \frac{z-4}{2}$
3. $\frac{x-2}{1} = \frac{y-1}{1} = \frac{z-4}{2}$
4. $\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.81

If $|\vec{a}| = 10$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = 12$, then value of $|\vec{a} \times \vec{b}|$ is :

1. 5
2. 10
3. 14
4. 16

Options 1. 1

2. 2
3. 3
4. 4

Q.82

For the matrix $A = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 2 & 3 \\ 1 & -2 & 1 \end{bmatrix}$, which of the following statements are \textbf{correct}?

- (A) The order of the matrix is 3×3
- (B) $|A| = 21$
- (C) $|\text{adj } A| = 225$
- (D) A is skew symmetric matrix

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only
2. (B), and (D) only
3. (A), and (C) only
4. (A), and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.83

If $|\vec{a}| = a$, then the value of $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$ is

1. a^2
2. $2a^2$
3. 0
4. $3a^2$

Options 1. 1

2. 2
3. 3
4. 4

Q.84

Let A, B, C be three events. If the probability of occurring exactly one out of A and B is $\frac{3}{5}$, exactly one out of B and C is $\frac{1}{5}$, exactly one out of C and A is $\frac{3}{5}$ and that of occurring of three events is $\frac{4}{25}$, then the probability of occurring at least

one of them is

1. $\frac{6}{25}$
2. $\frac{27}{50}$
3. $\frac{43}{50}$
4. $\frac{23}{25}$

Options

1. 1
2. 2
3. 3
4. 4

Q.85

The particular solution of the differential equation $\frac{dy}{dx} = e^{x^2/2} + xy$, when $x = 0, y = 1$, is

1. $y = e^{-x^2/2} + 1$
2. $y = (x + 1)e^{x^2/2}$
3. $y = xe^{x^2/2} + 1$
4. $y = (x - 1)e^{x^2/2}$

Options

1. 1
2. 2
3. 3
4. 4